

SOURAV TRIPATHY

+91 9337191375 · Bhubaneswar, India · lipuntripathy74@gmail.com

[linkedin.com/in/sourav-tripathy-astrophile](https://www.linkedin.com/in/sourav-tripathy-astrophile) · github.com/Sourav-Tripathy · souravtripathy.com

EDUCATION

B.Tech, Electronics and Telecommunication Engineering 2020 – 2024

IGIT Sarang, Odisha | *Final Year Project: Sink-hole Attack Detection using VANET Simulation Data*

SKILLS

Python, Supervised/Unsupervised Learning, NLP, Neural Networks (PyTorch, TensorFlow), Transformers, Reinforcement Learning (Policy Gradients, Value-based), LLMs, RAG, Context Optimization, Vector Search, Graph-RAG, PEFT (LoRA, QLoRA), SFT, CoT, LangGraph, CrewAI, Multi-Agent Orchestration, OpenTelemetry, Agent Tracing & Evals, FastAPI, WebSocket, Async Programming, MongoDB, Neo4j, Git, Linux, CI/CD

EXPERIENCE

Software Engineer (ML) | Cognitive View, Bhubaneswar, Odisha Aug 2024 – Present

- Deployed an LLM-powered research assistant over structured/unstructured data, reducing sales research turnaround from 1 day to under 2 hours — a 90% reduction in research friction.
- Built real-time multi-agent workflows (LangGraph + WebSockets) enabling parallel reasoning, achieving 3× faster LLM interaction across multi-step tasks.
- Architected a hybrid retrieval system (MongoDB vector search + Neo4j graph queries) with semantic routing, cutting average RAG latency from 12–15s to 4–6s across 1,000+ test queries.
- Designed an async FastAPI backend with persistent session memory and modular RAG orchestration, scaling to 500+ concurrent sessions without performance degradation.
- Building end-to-end Evals and Observability infrastructure for multi-agent systems using OpenTelemetry protocols — covering trace collection, span-level metrics, and agent behavior analysis.

PROJECTS

PersonaPilot github.com/Sourav-Tripathy/personapilot — Voice-agent roleplay simulation tool for sales training. Multiple LLM-backed persona agents (FastAPI + MongoDB + Next.js) conduct outbound calls via Bolna, capturing live transcripts, frustration scores, and friction markers for coaching review.

CommitGen github.com/Sourav-Tripathy/commitgen — CLI tool that auto-generates Conventional Commits messages from git diffs using local LLMs (Ollama) or cloud providers (OpenRouter). Handles large diffs via smart chunking; published as commitgen-ai on PyPI.

Alignment Study github.com/Sourav-Tripathy/Alignment-Study — Empirical experiments on LLM behavioral priors. Exp 1 (Split or Steal): cooperative alignment collapses under adversarial history. Exp 2 (Newcomb's Problem): models hold a strong EDT prior, dynamically shifting to Causal reasoning when predictor accuracy framing changes.

Embedding Crosslinguality Study github.com/Sourav-Tripathy/embedding-crosslinguality-study — Analyzed multilingual embeddings across 6 languages using LLaMA + multilingual-MiniLM. Measured cosine similarity and vector order consistency across translated essays and words.

LoRA Fine-Tuning of Qwen-1.5 — Fine-tuned Qwen-1.5B on a custom instruction set via PEFT LoRA. Demonstrated improved relevance and domain adaptation on targeted prompts.

WRITING & RESEARCH INTERESTS

- **Silicon & Soul** (siliconandsoul.substack.com) — Technical writing at the intersection of ML engineering and inference systems. Recent topics: Distributed Inference constraints, Python Webservers analysis, Continual Learning, KL-divergence
- **Open-Source Contribution - Otari/Mozilla-ai** (<https://github.com/mozilla-ai/otari>)
- Active study in LLM alignment, agent evals, observability infrastructure, and retrieval system design.